

Kubernetes Cluster Setup SOP (Using kubeadm & containerd)

Objective

- 1 To set up a Kubernetes cluster using kubeadm, with containerd as the container runtime.

Prerequisites

- 1 Linux system (Debian/Ubuntu-based)
- 2 Root or sudo access
- 3 Minimum: 2 CPUs, 2GB RAM
- 4 Network connectivity between nodes
- 5 Unique hostname for each node

System Configuration (All Nodes)

- 1 Enable IP Forwarding
- 2

```
sudo -i
cat <<EOF | tee /etc/sysctl.d/k8s.conf
net.ipv4.ip_forward = 1
EOF
sysctl --system
sysctl net.ipv4.ip_forward
```
- 3 Disable Swap
- 4

```
swapoff -a
sed -i '/ swap / s/^#/' /etc/fstab
```

Install Container Runtime (containerd)

- 1

```
apt update && apt install -y ca-certificates curl
```
- 2

```
install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/debian/gpg -o /etc/apt/keyrings/docker.asc
chmod a+r /etc/apt/keyrings/docker.asc
```
- 3

```
tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/debian
Suites: $(. /etc/os-release && echo "$VERSION_CODENAME")
Components: stable
Architectures: $(dpkg --print-architecture)
Signed-By: /etc/apt/keyrings/docker.asc
EOF
```
- 4

```
apt update && apt install -y containerd.io
```
- 5

```
containerd config default > /etc/containerd/config.toml
# Edit file and set:
SystemdCgroup = true
systemctl restart containerd
systemctl enable containerd
```

Install Kubernetes Components

- 1

```
apt-get update
```

```
2 apt-get install -y apt-transport-https ca-certificates curl gpg
3 curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.33/deb/Release.key | gpg --dearmor -o /etc/apt/keyrings/
4 echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.
5 apt-get update && apt-get install -y kubelet kubeadm kubectl
6 apt-mark hold kubelet kubeadm kubectl
```

Initialize Control Plane (Master Node Only)

```
1 kubeadm init --apiserver-advertise-address=<MASTER_IP> --pod-network-cidr=10.244.0.0/16 --cri-socket uni
```

Configure kubectl (Non-root User)

```
1 exit
  mkdir -p $HOME/.kube
  sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
  sudo chown $(id -u):$(id -g) $HOME/.kube/config
```

Join Worker Nodes

```
1 kubeadm join <MASTER_IP>:6443 --token <TOKEN> --discovery-token-ca-cert-hash sha256:<HASH>
```

Install Pod Network (CNI)

```
1 kubectl apply -f https://reweave.azurewebsites.net/k8s/v1.33/net.yaml
```

Verification

```
1 kubectl get nodes
2 kubectl get pods -A
```

Troubleshooting Tips

```
1 systemctl status containerd
2 journalctl -u kubelet -f
```

Notes / Best Practices

- 1 Ensure required ports are open
- 2 Use static IPs for production
- 3 Secure cluster with RBAC and network policies